

SEMINAR III · WEEK 8 · FOUNDATIONS

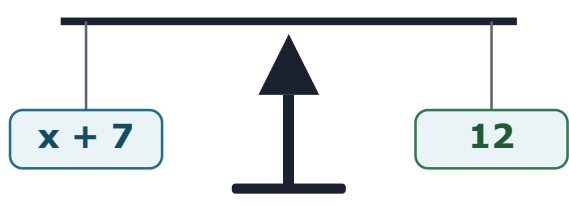
Equations

An equation is a balance. Whatever you do to one side, you do to the other — that single idea is the key to everything this week.

SESSION A

One-Step and Two-Step Equations

WHAT IS AN EQUATION?



Both sides stay level. Whatever operation you apply to one side, apply the same to the other — or the equation isn't balanced anymore.

INVERSE OPERATIONS

Undo what's currently being done to the variable.

Addition ↔ Subtraction

Multiplication ↔ Division

To solve $x + 7 = 12$, subtract 7 — not multiply by something. Use the inverse of what's actually there.

SOLVING ONE-STEP EQUATIONS

$x + 9 = 15 \rightarrow$ subtract 9 $\rightarrow x = 6$
 $x - 4 = 10 \rightarrow$ add 4 $\rightarrow x = 14$
 $4x = 28 \rightarrow$ divide by 4 $\rightarrow x = 7$
 $x \div 3 = 6 \rightarrow$ multiply by 3 $\rightarrow x = 18$

SOLVING TWO-STEP EQUATIONS

Undo in reverse order.

$3x + 5 = 20$: subtract 5 $\rightarrow 3x = 15$. Divide by 3 $\rightarrow \mathbf{x = 5}$.

Addition/subtraction first, then multiplication/division — not the order the equation was built in. A negative coefficient doesn't need a different method, just the Week 3 sign rules when you divide.

BEFORE WE MOVE ON

1. Solve a two-step equation
2. Solve one with a negative coefficient
3. Identify the correct first step for a given equation

SESSION B

Combine/Distribute First, Then Check & Apply

BEFORE YOU SOLVE

Combine first: $2x+3x+4=19 \rightarrow 5x+4=19 \rightarrow 5x=15 \rightarrow x=3$
Distribute first: $3(x+2)=21 \rightarrow 3x+6=21 \rightarrow 3x=15 \rightarrow x=5$

Combining like terms doesn't change what the equation means or its solution. Distributing first is more reliable than working around the parentheses, especially with more terms involved.

CHECKING YOUR SOLUTION

Checking isn't re-solving.

Substitute your answer back into the *original* equation and confirm both sides come out equal. Once x is found, interpret it in context — with the right units — before calling it the final answer. And a solution doesn't have to be a whole number.

EQUATIONS IN REAL LIFE

A rental costs \$40 plus \$15 per day. Total cost was \$115. How many days?

$$15d + 40 = 115 \rightarrow 15d = 75 \rightarrow \mathbf{d = 5 \text{ days}}$$

Build the equation from the scenario, solve it, then say what the answer means.

BEFORE WE MOVE ON

Write an equation from a scenario, solve it, and state the answer with correct units and context.

Then: mixed practice, error analysis, Week 8 check, reflection.

TRANSITION TO WEEK 9

Quarter 1 Consolidation

Bringing the quarter's number sense and early algebra skills together before Quarter 2.